

- 1 (a) Define velocity.

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 [1]

- (b) In an experiment, two objects A and B are released from the side of a building, as shown in Fig. 1.1.

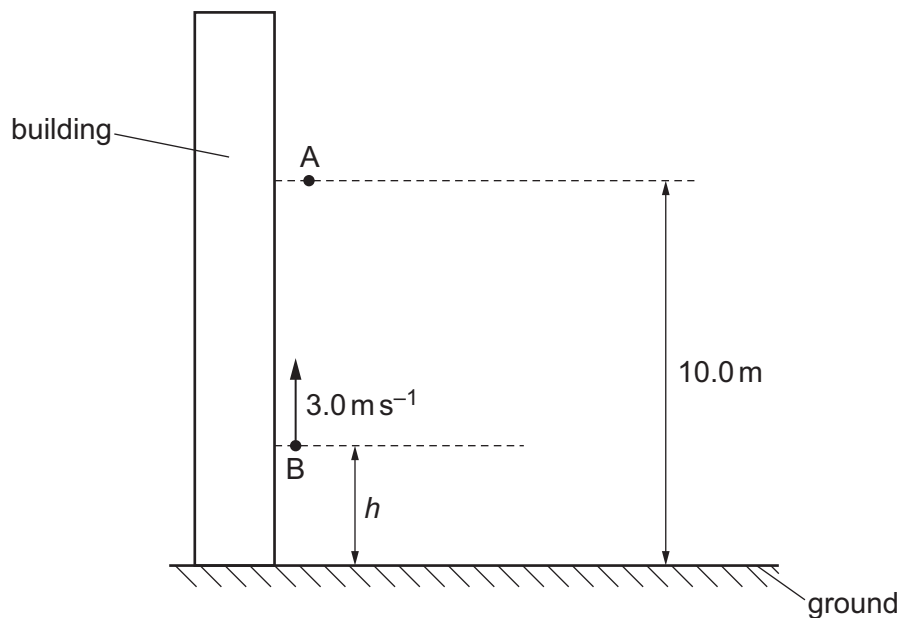


Fig. 1.1 (not to scale)

Object A is released from rest at a height of 10.0 m above horizontal ground.
 Object B is released with an initial upward velocity of 3.0 ms^{-1} at a height h above the ground.
 Both objects take the same time to reach the ground and they do not collide with each other.
 Air resistance is negligible.

Calculate h .

$h = \dots\dots\dots \text{ m [3]}$



- (c) In a second experiment, object B is released from the same height as in (b) but with a speed of 6.0 ms^{-1} at an angle of 60° to the vertical, as shown in Fig. 1.2.

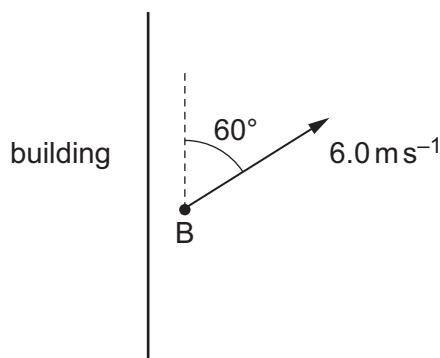


Fig. 1.2

- (i) State and explain whether the time taken for object B to reach the ground is less than, the same as, or greater than the time taken in the first experiment.

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 [2]

- (ii) By considering energy, state and explain whether the speed at which object B reaches the ground is less than, the same as, or greater than in the first experiment.

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 [2]

[Total: 8]

